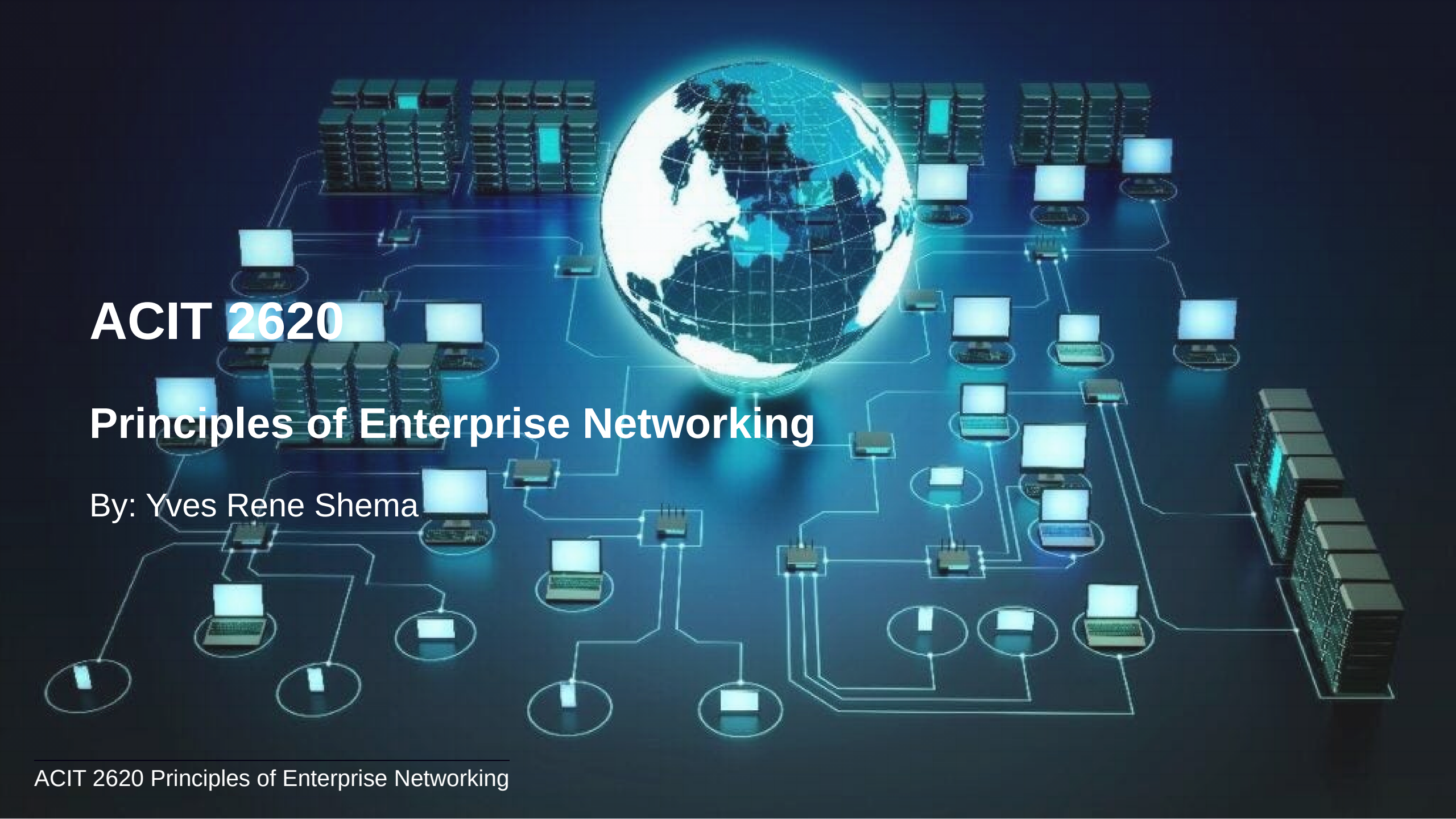


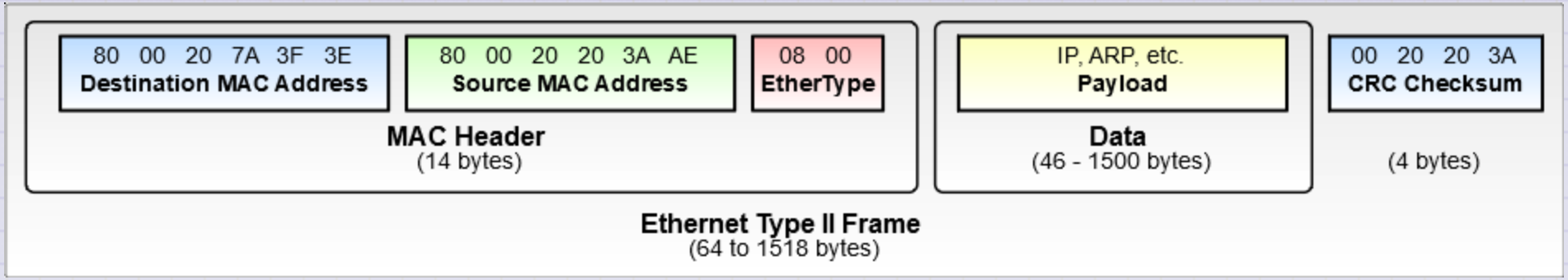
ACIT 2620

Principles of Enterprise Networking

By: Yves Rene Shema

Ethernet

Frames



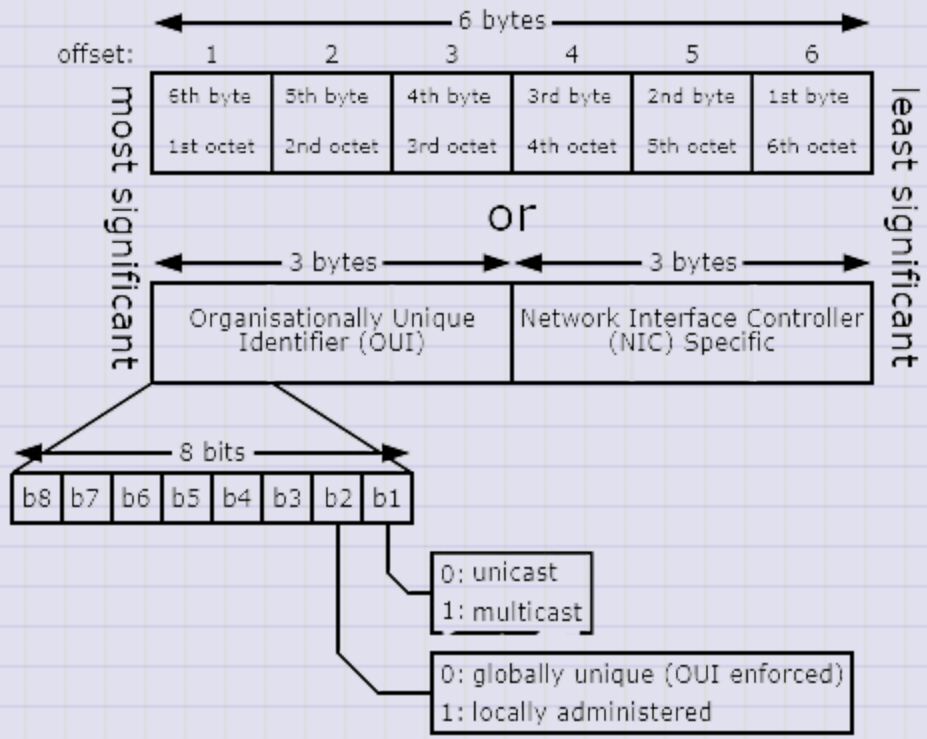
- Preamble
  - Marks beginning of entire frame
- Start of Frame Delimiter (SFD)
  - Indicates beginning of addressing fields
- Destination Address
  - Contains destination node address

- Source Address
  - contains address of sender node
- Length (LEN)
  - indicates length of data/payload
- Data (payload)
  - contains data, or segmented part of data, transmitted from originating node

- Pad
  - Used to increase size of the frame to its minimum size requirement of 46 bytes
- Frame Check Sequence
  - provides algorithm to determine whether data were correctly received
  - most commonly used algorithm is Cyclic Redundancy Check (CRC)

Ethernet Addressing

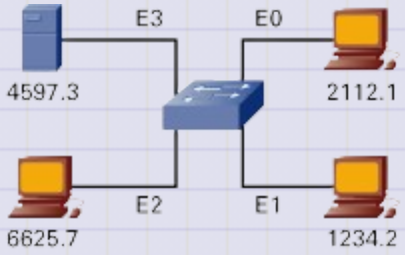
- MAC address: Media Access Control (MAC) sub-layer
- 48 Bits
- Number uniquely defining a network node
- Generally rendered as Hex: `00:1e:33:ba:87:c1`
- Doesn't contain any data regarding network location –just an ID


6 bytes
offset:
1
2
3
4
5
6
6th byte
1st octet
5th byte
2nd octet
4th byte
3rd octet
3rd byte
4th octet
2nd byte
5th octet
1st byte
6th octet
most significant
least significant
or
3 bytes
3 bytes
Organisationally Unique Identifier (OUI)
Network Interface Controller (NIC) Specific
8 bits
b8
b7
b6
b5
b4
b3
b2
b1
0: unicast
1: multicast
0: globally unique (OUI enforced)
1: locally administered

- first three bytes
  - Either Manufacturer hard coded
  - Or Reserved Addresses (common ones)
    - Broadcast Address `FF:FF:FF:FF:FF:FF`
    - Spanning Tree Multicast: `01:80:C2:00:00:00`
    - IANA reserves all address starting with `00:00:5E` see Ethernet Numbers(this includes IPv4 multicast -and inserts the low 23 Bits of the multicast IPv4 Address into the Ethernet Address)
    - `33:33:XX` is reserved for IPv6 Multicast

Switching

- Making Forwarding decisions
  - Transparent bridging



Time 0

Port	Address

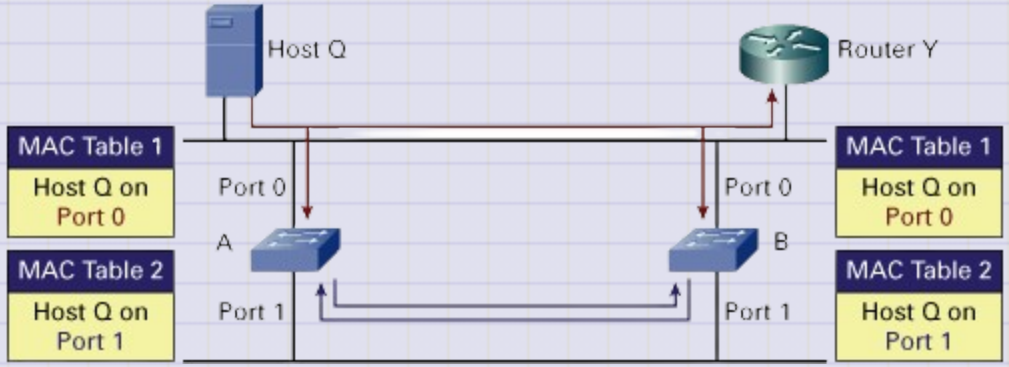
Time 1

Port	Address
E2	6625.7

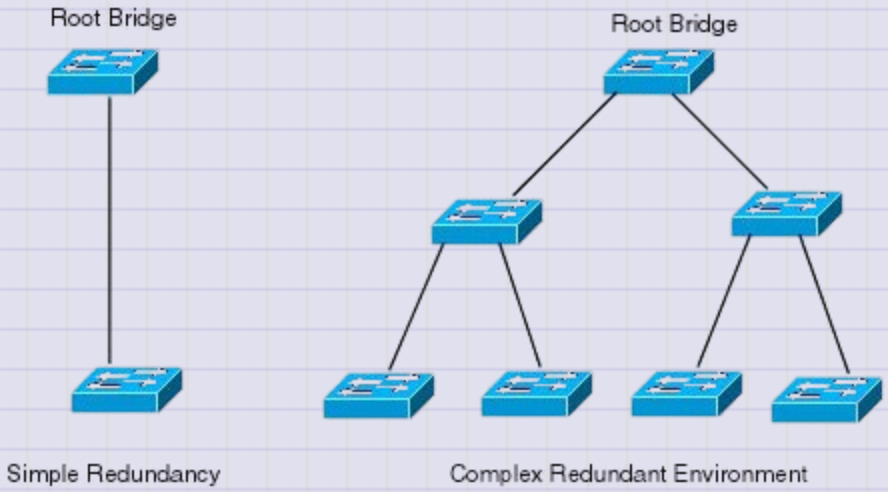
Time 2

Port	Address
E2	6625.7
E0	2112.1

Broadcast Loop and STP

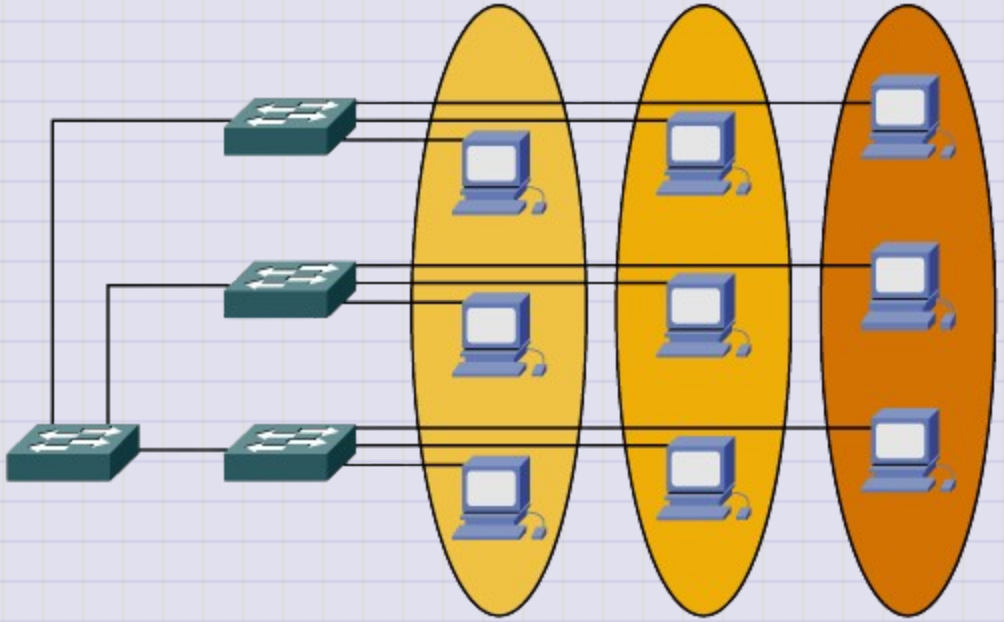


Spanning Tree Protocol



VLANs

- Virtual Local Area Networks
- A logical network within a physical network
- Achieved by grouping some of the switch ethernet ports into a logical broadcast domain
- Can span multiple switches

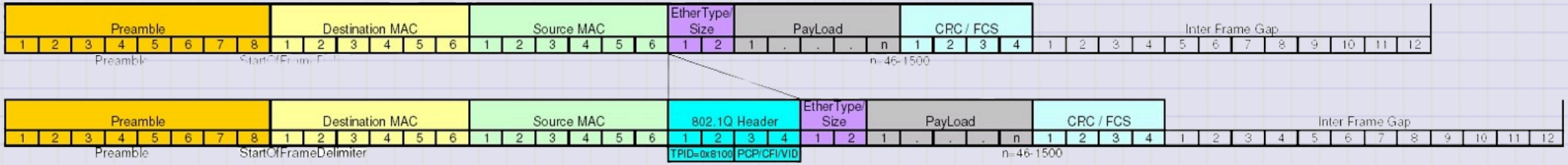


VLAN port types

- Access ports
  - assigned VLAN ID
  - for connecting end hosts/nodes
  - nodes connected to ports with same VLAN ID are in the same broadcast domain

- Trunk ports
  - typically for switch to switch or switch to router connection
  - carry "tagged" frames, i.e modified ethernet frames with VLAN markers

Tagged frames



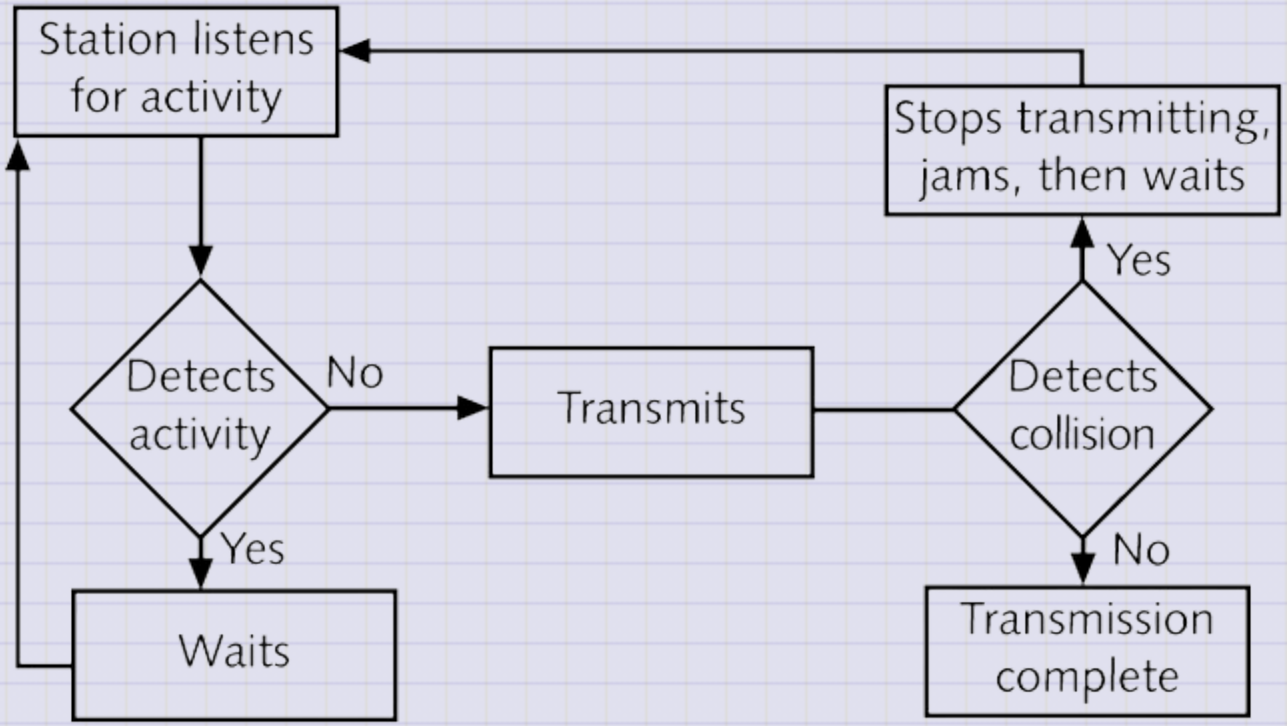
- 4-byte tag header inserted between Source MAC and EtherType fields
  - 2-byte tag protocol identifier (TPID)
    - a fixed value of 0x8100 that indicates the frame carries tag information.

- 2-byte tag control information (TCI)
  - Three-bit user priority (used to prioritize traffic)
  - Drop Eligible Indicator (DEI) (in congestion is frame “dropable”)
  - Twelve-bit VLAN identifier (VID)-Uniquely identifies the VLAN to which the frame belongs

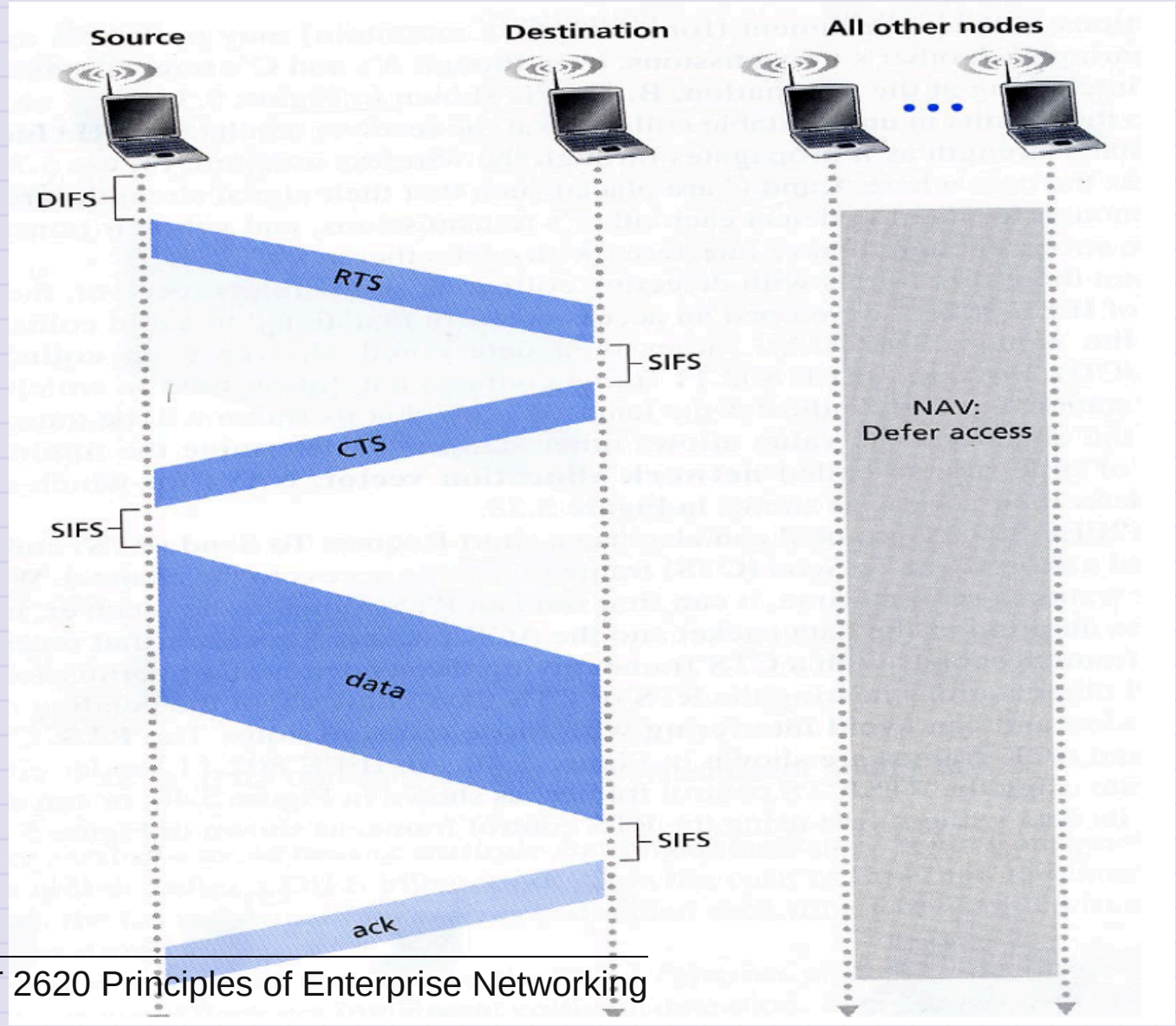
Link Access Methods

- Manage shared medium access contention (collision)
- Two methods:
  - CSMA/CD: for wired Ethernet
  - CSMA/CA: for wireless Ethernet

CSMA/CD


Station listens for activity
Detects activity
No
Yes
Transmits
Waits
Detects collision
Yes
No
Stops transmitting, jams, then waits
Transmission complete

CSMA/CA


Source
Destination
All other nodes
DIFS
RTS
SIFS
CTS
SIFS
NAV:
Defer access
data
SIFS
ack